

MamaStoria Infrastructure Configuration

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MamaStoria

Cloud Infrastructure Configuration

1. Overview

This document contains the current cloud infrastructure configuration for MamaStoria application deployed on Google Cloud Platform (GCP). The application uses Cloud Run for containerized services, Cloud SQL for database, and Cloud Storage for media files.

2. Cloud Run Services

2.1 nanobanana-backend (Main API)

Parameter	Value
Service Name	nanobanana-backend
Region	asia-southeast2 (Jakarta)
URL	https://nanobanana-backend-...run.app
CPU	4 vCPU
Memory	4 GB
Concurrency	100 requests/instance
Min Instances	2 (always warm)
Max Instances	50
Timeout	300 seconds
CPU Throttling	Disabled

Capacity: $50 \times 100 = 5,000$ concurrent requests

2.2 nanobanana-worker (Background Jobs)

Parameter	Value
Service Name	nanobanana-worker
Region	asia-southeast2 (Jakarta)
CPU	2 vCPU
Memory	4 GB
Concurrency	1 request/instance
Max Instances	50

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Purpose	Long-running comic generation
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2.3 smart-crop-worker (OpenCV Cropping)

Parameter	Value
Service Name	smart-crop-worker
Type	Cloud Functions Gen2
Region	asia-southeast2 (Jakarta)
CPU	1 vCPU
Memory	2 GB
Concurrency	10 requests/instance
Max Instances	100

Capacity: 100 x 10 = 1,000 concurrent crops

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3. Database Configuration

Cloud SQL (PostgreSQL)

Parameter	Value
Instance Name	cloudsql-nanobanana-dev
Region	asia-southeast2
Type	PostgreSQL
Connection	Cloud SQL Connector

4. Storage Configuration

Google Cloud Storage

Parameter	Value
Bucket Name	mamastoria-storage
Region	asia-southeast2
Access	Public (comic assets)

Storage Paths:

comics/panels/{comic_id}/	- Panel images
comics/videos/{comic_id}/	- Generated videos
comics/grids/{comic_id}/	- Full grid images
users/avatars/	- User profile pictures

5. External Services

Google Cloud Text-to-Speech

Parameter	Value
Voice	id-ID-Wavenet-A
Language	Indonesian (id-ID)
Gender	Female
Speaking Rate	0.9

Vertex AI (Image Generation)

Parameter	Value
Location	us-central1
Model	imagen-3.0-generate-002

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6. Capacity Summary

For 1,000 Concurrent Users

Service	Capacity
API Requests	5,000 concurrent - SUFFICIENT
Panel Cropping	1,000 concurrent - SUFFICIENT
Comic Generation	50 concurrent - SUFFICIENT

Response Times (P95)

Endpoint	Latency
List Comics	< 150ms
Comic Detail	< 130ms
Comic Panels	< 120ms
List Styles	< 250ms
List Genres	< 125ms

7. Cost Estimation (Monthly)

Service	Est. Cost
Cloud Run (backend, 2 min instances)	~\$50/month
Cloud Run (worker, pay per use)	~\$20/month
Cloud Functions (pay per use)	~\$10/month
Cloud SQL (db-f1-micro)	~\$10/month
Cloud Storage (~50GB)	~\$5/month
Vertex AI (per image)	Variable
Text-to-Speech (per character)	Variable

Base Infrastructure: ~\$95/month (excluding AI usage)

8. Project Information

Field	Value
Project ID	nanobananacomic-482111
Primary Region	asia-southeast2 (Jakarta)
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